



SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

Continuous Assessment Test-1, Fall Semester 2021-2022

CSE3013 – Artificial Intelligence

Slot: C1+TC1

Maximum Marks: 30

General Instructions (if any): Answer ALL Questions

S.No	Question	Ma x Ma rks	CO	BL																
1	Pick an Artificial Intelligence system that has appeared in the news or on TV recently and work out PAGE (i.e. Percepts, Actions, Goals, Environment) descriptions for it. How do these relate to their PEAS . (i.e. Performance, Environment, Actuators, Sensors) with description of the task environment and characterize it in terms of the properties.	10	CO1	L1																
2	The graph in the figure below shows the state space of a hypothetical search problem. States are denoted by letters, and the cost of each action is indicated on the corresponding edge. Note that actions are not reversible, since the graph is oriented. The table next to the state space shows the value of some admissible heuristic function, considering G as the goal state (it is easy to verify that such an heuristic never overestimates the true, minimum path cost from any given state to the goal state G). state space heuristic function (goal state: G) <table><tr><th>S</th><th>A</th><th>B</th><th>C</th><th>D</th><th>E</th><th>F</th><th>G</th></tr><tr><td>6</td><td>4</td><td>5</td><td>2</td><td>2</td><td>8</td><td>4</td><td>0</td></tr></table> Considering S as the initial state, solve the above search problem using: i) breadth-first search ii) depth-first search iii) A* search with the heuristic above When drawing the search tree, you should clearly indicate: the order of expansion of each node (e.g., by numbering the expanded nodes according to the order of their expansion); the action corresponding to each edge of the tree; the state, the path cost and the value of the heuristic of each node. In the case of depth-first search you should also indicate which nodes of the search tree are simultaneously stored in computer memory (in the case of a computer implementation) during the search process	S	A	B	C	D	E	F	G	6	4	5	2	2	8	4	0	10	CO4	L3
S	A	B	C	D	E	F	G													
6	4	5	2	2	8	4	0													
3	Write the pseudo code for Alpha- Beta Pruning. Apply both minimax algorithm and alpha beta pruning to the following graph. What are the nodes that are to be pruned?	10	CO4	L3																

(2+3+5)

Consider the Minimax Search Tree Expanded 2-ply, the square nodes are MAX nodes and the circular nodes are MIN nodes.

□ Max Nodes

○ Min Nodes

